

```

public class lab3one {
    public static void main(String[] args) {
        int n = 10000;

        for(int i = 1; i < 4; i++){
            System.out.println(n);
        }
    }
}

```

```

C:\xampp\htdocs\BUSTAMANTE> cmd /C "C:\Program Files\Java\jdk-7.0.75\bin\java.exe -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3one.jar -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3one.jar -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3one.jar"
10000
10000
10000

C:\xampp\htdocs\BUSTAMANTE>

```

```

import java.util.Scanner;

public class lab3two {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = scanner.nextInt();

        for(int i = 1; i < n + 1; i++){
            for(int j = 1; j <= Math.log(n) + 1 ; j++){
                System.out.printf("(%d, %d)%n", i, j);
            }
        }
        scanner.close();
    }
}

```

```

C:\xampp\htdocs\BUSTAMANTE> cmd /C "C:\Program Files\Java\jdk-7.0.75\bin\java.exe -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3two.jar -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3two.jar -Djava.class.path=C:\xampp\htdocs\BUSTAMANTE\lab3two.jar"
Enter a number: 3
(1, 1)
(1, 2)
(2, 1)
(2, 2)
(3, 1)
(3, 2)

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```

```

public class lab3three {
    public static void main(String[] args) {
        int[] arr = {1, 3, 5, 7, 9, 11, 13, 15};
        int target = 7;
        int result = binarySearch(arr, target);

        System.out.println(result);
    }

    public static int binarySearch(int[] arr, int target) {
        int low = 0;
        int high = arr.length - 1;

        while (low <= high) {
            int mid = (high + low) / 2;
            if (arr[mid] < target) {
                low = mid + 1 ;
            } else if (arr[mid] > target) {
                high = mid - 1;
            } else {
                return mid;
            }
        }
        return -1;
    }
}

```

```

C:\xampp\htdocs\BUSTAMANTE> cmd
7cee0675b3ed02\redhat.java\jdt_v
3

C:\xampp\htdocs\BUSTAMANTE>

```

```

import java.util.Scanner;

public class lab3four {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = scanner.nextInt();

        for(int i = 1; i < n + 1; i++){
            for(int j = 1; j < n + 1; j++){
                for(int k = 1; k < n + 1; k++){
                    System.out.printf("(%d, %d, %d)%n", i, j, k);
                }
            }
        }
        scanner.close();
    }
}

```

```

C:\xampp\htdocs\BUSTAMANTE> cmd /C ""
7cee0675b3ed02\redhat.java\jdt_ws\BUS
Enter a number: 2
(1, 1, 1)
(1, 1, 2)
(1, 2, 1)
(1, 2, 2)
(2, 1, 1)
(2, 1, 2)
(2, 2, 1)
(2, 2, 2)

C:\xampp\htdocs\BUSTAMANTE>

```

```

import java.util.Scanner;

public class lab3five {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = scanner.nextInt();
        for (int i = 0; i < n; i++){
            System.out.println(Integer.toBinaryString(i + 1));
        }
        scanner.close();
    }
}

```

```

C:\xampp\htdocs\BUSTAMANTE> cmd
7cee0675b3ed02\redhat.java\jdt_w
Enter a number: 4
1
10
11
100

C:\xampp\htdocs\BUSTAMANTE>

```

```

public class lab3six {
    public static void main(String[] args) {
        int n = 10;
        int num = 0;

        for (int i = 1; i <= n; i++) {
            if (num == 0) {
                System.out.println(num);
                num = 1;
            }
            num = num * i;
            System.out.println(num);
        }
    }
}

```

```
C:\xampp\htdocs\BUSTAMANTE> cmd /C ""0  
7cee0675b3ed02\redhat.java\jdt_ws\BUST  
0  
1  
2  
6  
24  
120  
720  
5040  
40320  
362880  
3628800  
  
C:\xampp\htdocs\BUSTAMANTE>
```